

Candidate Theory Landscapes for Feedback Control

Paper notation: $x = n/N$, b , q_c , β , θ , h_c , h_e , γ , T_π , J_c , and η_{th} .

1. Baseline landscape: $T_\pi(x, b)$

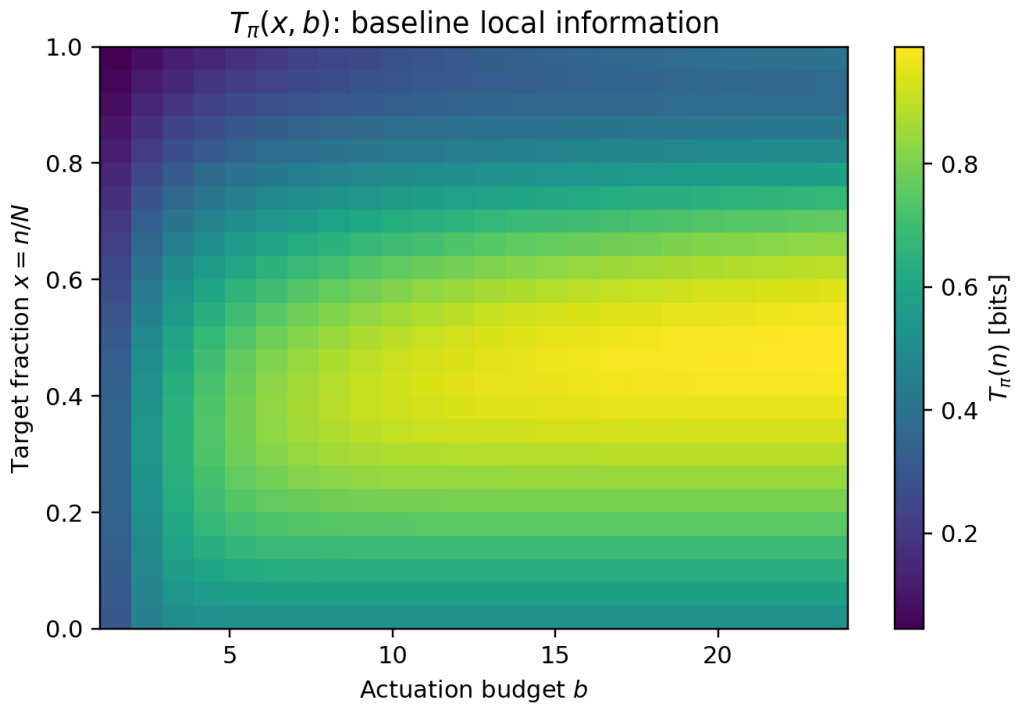
For a fixed pre-actuation state n , the paper defines

$$T_\pi(n) = I(U; m \mid n),$$

where $U \in \{0, 1\}$ is the controller action and m is the post-actuation population count. The feedback action probability is

$$a_n = P(U = 1 \mid n) = \sum_y S(y \mid n) \sigma \left[\beta \left(\theta - \frac{y}{q_c} \right) \right].$$

The first landscape keeps the controller policy fixed and changes only the current population state $x = n/N$ and the actuation budget b .



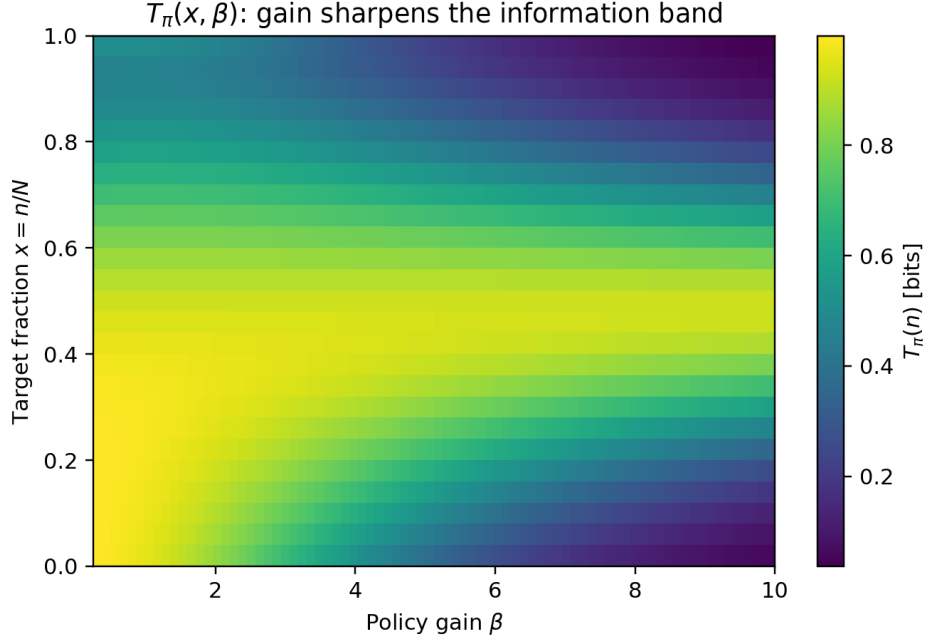
This is the cleanest summary of the operational information channel. Increasing b separates the ADVOCATE and NoOp next-state kernels, but T_π remains small when the action itself is nearly deterministic. The resulting ridge therefore reflects two ingredients at once: *action uncertainty* and *dynamical separability*.

2. Policy geometry: $T_\pi(x, \beta)$ and $T_\pi(x, \theta)$

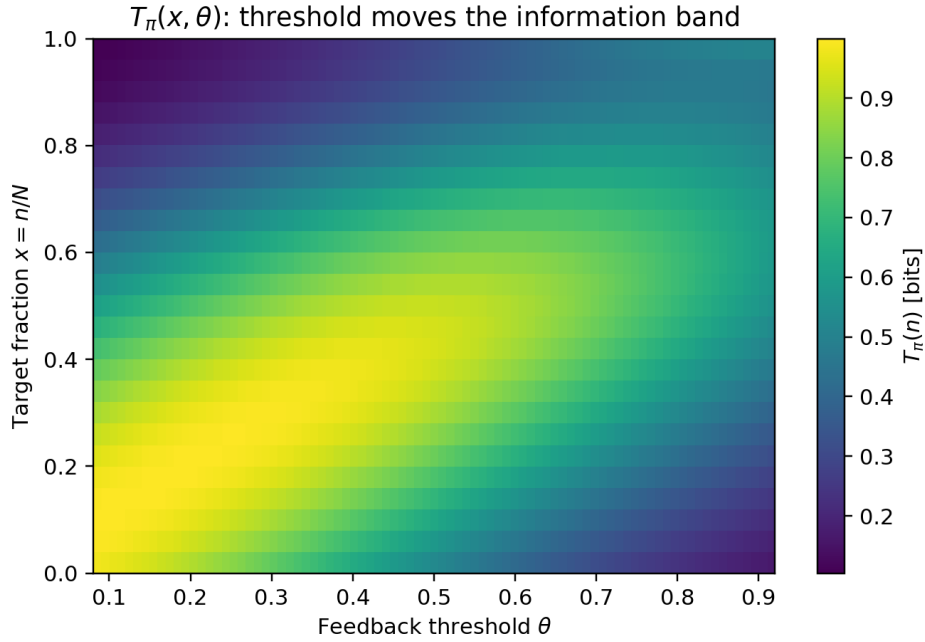
The policy is

$$P(U = 1 \mid Y = y) = \sigma \left[\beta \left(\theta - \frac{y}{q_c} \right) \right].$$

This makes the roles of β and θ geometrically distinct.



At fixed $b = 12$, increasing β makes the policy more switch-like. The controller becomes more deterministic away from its decision boundary, so the informative region contracts rather than simply growing everywhere. Thus β primarily controls the *width and sharpness* of the information band.



Changing θ instead moves the switching region through population state space. Roughly, the informative band tracks the states in which the sampled target fraction Y/q_c is close enough to θ that both actions remain possible. Hence θ controls *where* feedback is deployed, whereas β controls *how sharply* that deployment changes.

3. Thermodynamic landscape: $\eta_{\text{th}}(h_c, h_e)$

The thermodynamic layer introduces the opposing load

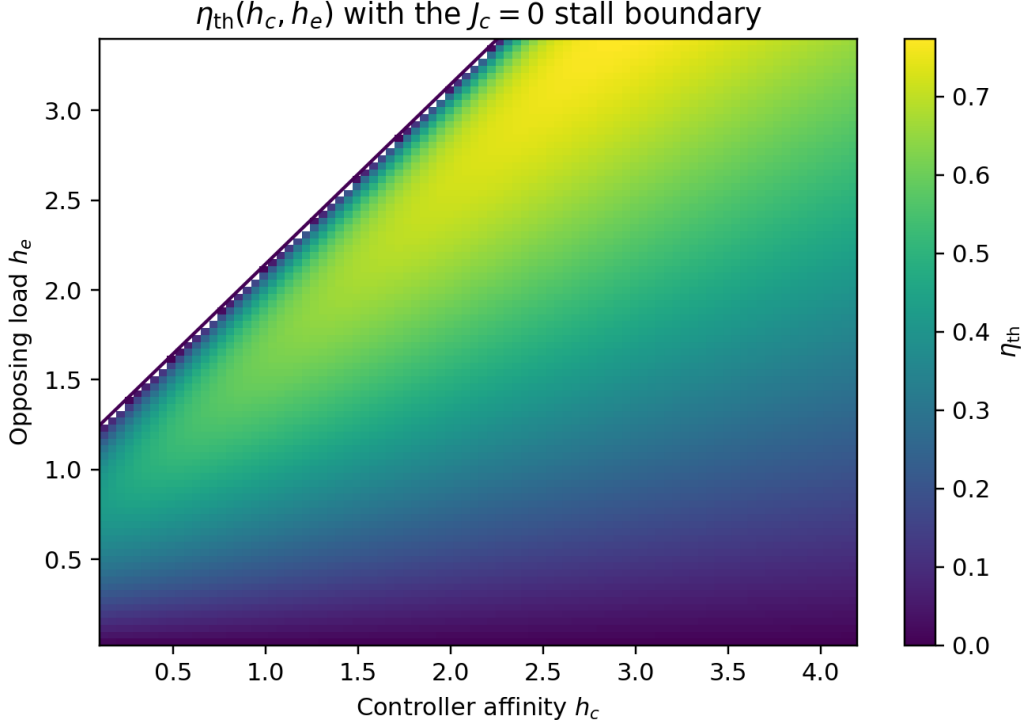
$$F = h_c - h_e, \quad J_c = N\Lambda_{b,\gamma} \sum_n p(n)a_n \left[\sigma(h_c - h_e) - \frac{n}{N} \right],$$

with

$$\Lambda_{b,\gamma} = 1 - \left(1 - \frac{\gamma}{N} \right)^b.$$

The finite-time efficiency is

$$\eta_{\text{th}} = \frac{h_e J_c}{h_c J_c + I(n; Y) + \Delta S_{\text{sys}}}, \quad J_c > 0, \quad h_e > 0.$$



The black contour is the stall boundary $J_c = 0$. Below the useful-operating side of this boundary the controller produces positive target-directed current; beyond it the opposing load overwhelms the controller. The efficiency is therefore not maximized by making either h_c or h_e arbitrarily large. The landscape exposes an operating region in which useful output against the load is large relative to controller work, sensing information, and finite-time system-entropy change.

For this illustrative exact theory plot we use the same ensemble as in the theory report:

$$p(n) = \text{Binomial}(N, 0.25), \quad N = 24, \quad b = 12, \quad q_c = 12, \quad \beta = 4, \quad \theta = 1/2, \quad \gamma = 0.75.$$

No empirical fit is used in this plot.

4. What these four diagrams add

The four landscapes answer four different questions:

- $T_\pi(x, b) :$ where does additional actuation make the action visible?
- $T_\pi(x, \beta) :$ how does feedback gain sharpen the information channel?
- $T_\pi(x, \theta) :$ how does the intervention threshold move that channel?
- $\eta_{\text{th}}(h_c, h_e) :$ where does the controller operate efficiently against a load?

They are complementary rather than interchangeable. The first three belong to the operational information layer: they describe the geometry of $T_\pi(n) = I(U; m \mid n)$. The fourth belongs to the thermodynamic layer and depends on the *sensing* information $I(n; Y)$, not on T_π .

For the paper, the strongest pair is likely

$$\boxed{T_\pi(x, b) \quad \text{and} \quad \eta_{\text{th}}(h_c, h_e),}$$

because they expose the two distinct bottlenecks of the controller: state-dependent information transfer and useful operation against an opposing load. The (x, β) and (x, θ) landscapes then give theory-driven experimental predictions. In particular, β should sharpen the empirical information ridge while θ should translate it through state space.

Baseline values for the information landscapes:

$$N = 24, \quad q_c = 12, \quad h_c = 2, \quad \beta = 4, \quad \theta = \frac{1}{2}, \quad \gamma = 1,$$

with $b = 12$ when b is not itself an axis.